



SECTION 8

INSTRUCTION FORMAT SUMMARY

This section contains a listing of the M68000 family instructions in binary format. It is listed in opcode order for the M68000 family instruction set.

8.1 INSTRUCTION FORMAT

The following paragraphs present a summary of the binary encoding fields.

8.1.1 Coprocessor ID Field

This field specifies which coprocessor in a system is to perform the operation. When using directly supported floating-point instructions for the MC68040, this field must be set to one.

8.1.2 Effective Address Field

This field specifies which addressing mode is to be used. For some operations, there are hardware-enforced restrictions on the available addressing modes allowed.

8.1.3 Register/Memory Field

This field is common to all arithmetic instructions. A zero in this field indicates a register-to-register operation, and a one indicates an < ea > -to-register operation.

8.1.4 Source Specifier Field

This field is common to all arithmetic instructions. The value of the register/memory (R/M) field affects this field's definition. If R/M = 0, specifies the source floating-point data register (FPDR). If R/M = 1, specifies the source operand data format.

- 000 — Long-Word Integer (L)
- 001 — Single-Precision Real (S)
- 010 — Extended-Precision Real (X)
- 011 — Packed-Decimal Real (P)
- 100 — Word Integer (W)
- 101 — Double-Precision Real (D)
- 110 — Byte Integer (B)

8.1.5 Destination Register Field

This field is common to all arithmetic instructions. It specifies the FPDR that that will be the destination. The results are always stored in this register.

8.1.6 Conditional Predicate Field

This field is common to all conditional instructions and specifies the conditional test that is to be evaluated. Table 8-1 shows the binary encodings for the conditional tests.

8.1.7 Shift and Rotate Instructions

The following paragraphs define the fields used with the shift and rotate instructions.

8.1.7.1 Count Register Field. If $i/r = 0$, this field contains the rotate (shift) count of 1 – 8 (a zero specifies 8). If $i/r = 1$, this field specifies a data register that contains the rotate (shift) count. The following shift and rotate fields are encoded as follows:

dr field

- 0 — Rotate (shift) Right
- 1 — Rotate (shift) Left

i/r field

- 0 — Immediate Rotate (shift) Count
- 1 — Register Rotate (shift) Count

8.1.7.2 Register Field. This field specifies a data register to be rotated (shifted).

Table 8-1. Conditional Predicate Field Encoding

Conditional Predicate	Mnemonic	Definition
000000	F	False
000001	EQ	Equal
000010	OGT	Ordered Greater Than
000011	OGE	Ordered Greater Than or Equal
000100	OLT	Ordered Less Than
000101	OLE	Ordered Less Than or Equal
000110	OGL	Ordered Greater Than or Less Than
000111	OR	Ordered
001000	UN	Unordered
001001	UEQ	Unordered or Equal
001010	UGT	Unordered or Greater Than
001011	UGE	Unordered or Greater Than or Equal
001100	ULT	Unordered or Less Than
001101	ULE	Unordered or Less Than or Equal
001110	NE	Not Equal
001111	T	True
010000	SF	Signaling False
010001	SEQ	Signaling Equal
010010	GT	Greater Than
010011	GE	Greater Than or Equal
010100	LT	Less Than
010101	LE	Less Than or Equal
010110	GL	Greater Than or Less Than
010111	GLE	Greater Than or Less Than or Equal
011000	NGLE	Not (Greater Than or Less Than or Equal)
011001	NGL	Not (Greater Than or Less Than)
011010	NLE	Not (Less Than or Equal)
011011	NLT	Not (Less Than)
011100	NGE	Not (Greater Than or Equal)
011101	NGT	Not (Greater Than)
011110	SNE	Signaling Not Equal
011111	ST	Signaling True

8.1.8 Size Field

This field specifies the size of the operation. The encoding is as follows:

- 00 — Byte Operation
- 01 — Word Operation
- 10 — Long Operation

8.1.9 Opcode Field

Refer to the applicable instruction descriptions for the encoding of this field in **Section 4 Integer Instructions**, **Section 5 Floating Point Instructions**, **Section 6 Supervisor (Privileged) Instructions**, and **Section 7 CPU32 Instructions**.

8.1.10 Address/Data Field

This field specifies the type of general register. The encoding is:

- 0 — Data Register
- 1 — Address Register

8.2 OPERATION CODE MAP

Table 8-2 lists the encoding for bits 15 – 12 and the operation performed.

Table 8-2. Operation Code Map

Bits 15 – 12	Operation
0000	Bit Manipulation/MOVEP/Immediate
0001	Move Byte
0010	Move Long
0011	Move Word
0100	Miscellaneous
0101	ADDQ/SUBQ/Scc/DBcc/TRAPcc
0110	Bcc/BSR/BRA
0111	MOVEQ
1000	OR/DIV/SBCD
1001	SUB/SUBX
1010	(Unassigned, Reserved)
1011	CMP/EOR
1100	AND/MUL/ABCD/EXG
1101	ADD/ADDX
1110	Shift/Rotate/Bit Field
1111	Coprocessor Interface/MC68040 and CPU32 Extensions

ORI to CCR

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	0	0	0	0	0	0	1	1	1	1	0	0
0	0	0	0	0	0	0	0	8-BIT BYTE DATA							

ORI to SR

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	0	0	0	0	0	1	1	1	1	1	0	0
16-BIT WORD DATA															

ORI

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	0	0	0	0	SIZE		EFFECTIVE ADDRESS					
										MODE			REGISTER		
16-BIT WORD DATA								8-BIT BYTE DATA							
32-BIT LONG DATA															

ANDI to CCR

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	0	0	1	0	0	0	1	1	1	1	0	0
0	0	0	0	0	0	0	0	8-BIT BYTE DATA							

ANDI to SR

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	0	0	1	0	0	1	1	1	1	1	0	0
16-BIT WORD DATA															

ANDI

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	0	0	0	0	SIZE		EFFECTIVE ADDRESS					
										MODE			REGISTER		
16-BIT WORD DATA								8-BIT BYTE DATA							
32-BIT LONG DATA															



SUBI

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0						
0	0	0	0	0	1	0	0	SIZE		EFFECTIVE ADDRESS											
										MODE		REGISTER									
16-BIT WORD DATA								8-BIT BYTE DATA													
32-BIT LONG DATA																					

RTM

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	0	1	1	0	1	1	0	0	D/A	REGISTER		

CALLM

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	0	1	1	0	1	1	EFFECTIVE ADDRESS					
										MODE		REGISTER			
0	0	0	0	0	0	0	0	ARGUMENT COUNT							

ADDI

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0						
0	0	0	0	0	1	1	0	SIZE		EFFECTIVE ADDRESS											
										MODE		REGISTER									
16-BIT WORD DATA								8-BIT BYTE DATA													
32-BIT LONG DATA																					

CMP2

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	0	SIZE		0	1	1	EFFECTIVE ADDRESS					
										MODE		REGISTER			
D/A	REGISTER			0	0	0	0	0	0	0	0	0	0	0	0

CHK2

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	0	SIZE		0	1	1	EFFECTIVE ADDRESS					
										MODE		REGISTER			
D/A	REGISTER			1	0	0	0	0	0	0	0	0	0	0	0

EORI to CCR

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	1	0	1	0	0	0	1	1	1	1	0	0
0	0	0	0	0	0	0	0	BYTE DATA (8 BITS)							

EORI to SR

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	1	0	1	0	0	1	1	1	1	1	0	0
16-BIT WORD DATA															

EORI

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	1	0	1	0	SIZE		EFFECTIVE ADDRESS					
										MODE			REGISTER		
16-BIT WORD DATA								16-BIT BYTE DATA							
32-BIT LONG DATA															

CMPI

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	1	1	0	0	SIZE		EFFECTIVE ADDRESS					
										MODE			REGISTER		
16-BIT WORD DATA								8-BIT BYTE DATA							
32-BIT LONG DATA															

BTST

BIT NUMBER STATIC, SPECIFIED AS IMMEDIATE DATA

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	1	0	0	0	0	0	EFFECTIVE ADDRESS					
										MODE			REGISTER		
0	0	0	0	0	0	0	0	BIT NUMBER							

BCHG

BIT NUMBER STATIC, SPECIFIED AS IMMEDIATE DATA

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	1	0	0	0	0	1	EFFECTIVE ADDRESS					
										MODE			REGISTER		
0	0	0	0	0	0	0	0	BIT NUMBER							

BCLR

BIT NUMBER STATIC, SPECIFIED AS IMMEDIATE DATA

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	1	0	0	0	1	0	EFFECTIVE ADDRESS					
											MODE		REGISTER		
0	0	0	0	0	0	0	0	BIT NUMBER							

BSET

BIT NUMBER STATIC, SPECIFIED AS IMMEDIATE DATA

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	1	0	0	0	1	1	EFFECTIVE ADDRESS					
										MODE		REGISTER			
0	0	0	0	0	0	0	BIT NUMBER								

MOVES

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	1	1	1	0	SIZE		EFFECTIVE ADDRESS					
				MODE		REGISTER									
A/D	REGISTER			dr	0	0	0	0	0	0	0	0	0	0	0

CAS2

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	1	SIZE		0	1	1	1	1	1	1	0	0
D/A1	Rn1			0	0	0	Du1			0	0	0	Dc1		
D/A2	Rn2			0	0	0	Du2			0	0	0	Dc2		

CAS

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	1	SIZE		0	1	1	EFFECTIVE ADDRESS					
										MODE			REGISTER		
0	0	0	0	0	0	0	Du			0	0	0	Dc		

BTST

BIT NUMBER DYNAMIC, SPECIFIED IN A REGISTER

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	REGISTER			1	0	0	EFFECTIVE ADDRESS					
				MODE		REGISTER									

BCHG

BIT NUMBER DYNAMIC, SPECIFIED IN A REGISTER															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	REGISTER			1	0	1	EFFECTIVE ADDRESS MODE REGISTER					

BCLR

BIT NUMBER DYNAMIC, SPECIFIED IN A REGISTER															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	REGISTER			1	1	0	EFFECTIVE ADDRESS MODE REGISTER					

BSET

BIT NUMBER DYNAMIC, SPECIFIED IN A REGISTER															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	REGISTER			1	1	1	EFFECTIVE ADDRESS MODE REGISTER					

MOVEP

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	DATA REGISTER			OPMODE			0	0	1	ADDRESS REGISTER		
16-BIT DISPLACEMENT															

MOVEA

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	SIZE		DESTINATION REGISTER			0	0	1	SOURCE MODE REGISTER					

MOVE

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	SIZE		DESTINATION REGISTER MODE						SOURCE MODE REGISTER					

MOVE from SR

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	1	0	0	0	0	0	0	1	1	SOURCE MODE REGISTER					



MOVE from CCR

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	1	0	0	0	0	1	0	1	1	EFFECTIVE ADDRESS					
										MODE	REGISTER				

NEGX

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	1	0	0	0	0	0	0	SIZE1		EFFECTIVE ADDRESS					
										MODE		REGISTER			

CLR

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	1	0	0	0	0	1	0	SIZE		EFFECTIVE ADDRESS					
									MODE		REGISTER				

MOVE to CCR

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	1	0	0	0	1	0	0	1	1	EFFECTIVE ADDRESS					
										MODE	REGISTER				

NEG

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	1	0	0	0	1	0	0	SIZE		EFFECTIVE ADDRESS					
									MODE		REGISTER				

NOT

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	1	0	0	0	1	1	0	SIZE		EFFECTIVE ADDRESS					
									MODE		REGISTER				

MOVE to SR

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	1	0	0	0	1	1	0	1	1	EFFECTIVE ADDRESS					
										MODE	REGISTER				

EXT, EXTB

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	1	0	0	1	0	0	OPMODE			0	0	0	REGISTER		

LINK

LONG															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	1	0	0	1	0	0	0	0	0	0	0	1	REGISTER		
HIGH-ORDER DISPLACEMENT															
LOW-ORDER DISPLACEMENT															

NBCD

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	1	0	0	1	0	0	0	0	0	EFFECTIVE ADDRESS					
										MODE		REGISTER			

SWAP

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	1	0	0	1	0	0	0	0	1	0	0	0	REGISTER		

BKPT

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	1	0	0	1	0	0	0	0	1	0	0	1	VECTOR		

PEA

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	1	0	0	1	0	0	0	0	1	EFFECTIVE ADDRESS					
										MODE		REGISTER			

BGND

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	1	0	0	1	0	1	0	1	1	1	1	1	0	1	0



ILLEGAL

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	1	0	0	1	0	1	0	1	1	1	1	1	1	0	0

TAS

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	1	0	0	1	0	1	0	1	1	EFFECTIVE ADDRESS					
										MODE			REGISTER		

TST

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	1	0	0	1	0	1	0	SIZE		EFFECTIVE ADDRESS					
										MODE			REGISTER		

MULU

LONG															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	1	0	0	1	1	0	0	0	0	EFFECTIVE ADDRESS					
										MODE			REGISTER		
0	REGISTER DI			0	SIZE	0	0	0	0	0	0	0	REGISTER Dh		

MULS

LONG															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	1	0	0	1	1	0	0	0	0	EFFECTIVE ADDRESS					
										MODE			REGISTER		
0	REGISTER DI			1	SIZE	0	0	0	0	0	0	0	REGISTER Dh		

DIVU, DIVUL

LONG															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	1	0	0	1	1	0	0	0	0	EFFECTIVE ADDRESS					
										MODE			REGISTER		
0	REGISTER Dq			0	SIZE	0	0	0	0	0	0	0	REGISTER Dr		

DIVS, DIVSL

LONG															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	1	0	0	1	1	0	0	0	1	EFFECTIVE ADDRESS					
								MODE				REGISTER			
0	REGISTER Dq			1	SIZE	0	0	0	0	0	0	0	REGISTER Dr		

TRAP

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	1	0	0	1	1	1	0	0	1	0	0	VECTOR			

LINK

WORD															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	1	0	0	1	1	1	0	0	1	0	1	0	REGISTER		
WORD DISPLACEMENT															

UNLK

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	1	0	0	1	1	1	0	0	1	0	1	1	REGISTER		

MOVE USP

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	1	0	0	1	1	1	0	0	1	1	0	dr	REGISTER		

RESET

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	1	0	0	1	1	1	0	0	1	1	1	0	0	0	0

NOP

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	1	0	0	1	1	1	0	0	1	1	1	0	0	0	1

STOP

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	1	0	0	1	1	1	0	0	1	1	1	0	0	1	0
IMMEDIATE DATA															

RTE

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	1	0	0	1	1	1	0	0	1	1	1	0	0	1	1

RTD

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	1	0	0	1	1	1	0	0	1	1	1	0	1	0	0
16-BIT DISPLACEMENT															

RTS

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	1	0	0	1	1	1	0	0	1	1	1	0	1	0	1

TRAPV

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	1	0	0	1	1	1	0	0	1	1	1	0	1	1	0

RTR

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	1	0	0	1	1	1	0	0	1	1	1	0	1	1	1

MOVEC

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	1	0	0	1	1	1	0	0	1	1	1	1	0	1	dr
A/D	REGISTER				CONTROL REGISTER										

JSR

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	1	0	0	1	1	1	0	1	0	EFFECTIVE ADDRESS					
										MODE		REGISTER			

JMP

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	1	0	0	1	1	1	0	1	1	EFFECTIVE ADDRESS					
										MODE		REGISTER			

MOVEM

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	1	0	0	1	dr	0	0	1	SIZE	EFFECTIVE ADDRESS					
										MODE		REGISTER			
REGISTER LIST MASK															

LEA

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	1	0	0	REGISTER			1	1	1	EFFECTIVE ADDRESS					
							MODE			REGISTER					

CHK

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	1	0	0	REGISTER			SIZE			0	EFFECTIVE ADDRESS				
										MODE		REGISTER			

ADDQ

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	1	0	1	DATA			0	SIZE		EFFECTIVE ADDRESS					
										MODE		REGISTER			

SUBQ

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	1	0	1	DATA			1	SIZE		EFFECTIVE ADDRESS					
										MODE		REGISTER			



DBcc

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	1	0	1	CONDITION				1	1	0	0	1	REGISTER		
16-BIT DISPLACEMENT															

TRAPcc

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	1	0	1	CONDITION				1	1	1	1	1	OPMODE		
OPTIONAL WORD															
OR LONG WORD															

Scc

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	1	0	1	CONDITION				1	1	EFFECTIVE ADDRESS					
									MODE			REGISTER			

BRA

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	1	1	0	0	0	0	0	8-BIT DISPLACEMENT							
16-BIT DISPLACEMENT IF 8-BIT DISPLACEMENT = \$00															
32-BIT DISPLACEMENT IF 8-BIT DISPLACEMENT = \$FF															

BSR

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	1	1	0	0	0	0	1	8-BIT DISPLACEMENT							
16-BIT DISPLACEMENT IF 8-BIT DISPLACEMENT = \$00															
32-BIT DISPLACEMENT IF 8-BIT DISPLACEMENT = \$FF															

Bcc

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	1	1	0	CONDITION				8-BIT DISPLACEMENT							
16-BIT DISPLACEMENT IF 8-BIT DISPLACEMENT = \$00															
32-BIT DISPLACEMENT IF 8-BIT DISPLACEMENT = \$FF															

MOVEQ

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	1	1	1	REGISTER			0	DATA							

DIVU, DIVUL

WORD															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	0	0	0	REGISTER			0	1	1	EFFECTIVE ADDRESS					
											MODE		REGISTER		

SBCD

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	0	0	0	REGISTER Dy/Ay			1	0	0	0	0	R/M	REGISTER Dx/Ax		

PACK

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	0	0	0	REGISTER Dy/Ay			1	0	1	0	0	R/M	REGISTER Dx/Ax		
16-BIT EXTENSION: ADJUSTMENT															

UNPK

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	0	0	0	REGISTER Dy/Ay			1	1	0	0	0	R/M	REGISTER Dx/Ax		
16-BIT EXTENSION: ADJUSTMENT															

DIVS, DIVSL

WORD															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	0	0	0	REGISTER			1	1	1	EFFECTIVE ADDRESS					
											MODE		REGISTER		

OR

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	0	0	0	REGISTER			OPMODE			EFFECTIVE ADDRESS					
											MODE		REGISTER		



SUBX

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	0	0	1	REGISTER Dy/Ay			1	SIZE		0	0	R/M	REGISTER Dx/Ax		

SUB

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	0	0	1	REGISTER			OPMODE			EFFECTIVE ADDRESS					
										MODE		REGISTER			

SUBA

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	0	0	1	REGISTER			OPMODE			EFFECTIVE ADDRESS					
										MODE		REGISTER			

CMPM

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	0	1	1	REGISTER Ax			1	SIZE		0	0	1	REGISTER Ay		

CMP

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	0	1	1	REGISTER			OPMODE			EFFECTIVE ADDRESS					
										MODE		REGISTER			

CMPPA

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	0	1	1	REGISTER			OPMODE			EFFECTIVE ADDRESS					
										MODE			REGISTER		

EOR

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	0	1	1	REGISTER			OPMODE			EFFECTIVE ADDRESS					
										MODE		REGISTER			

MULU

WORD															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	0	0	REGISTER			0	1	1	EFFECTIVE ADDRESS					
										MODE			REGISTER		

ABCD

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	0	0	REGISTER Rx			1	0	0	0	0	R/M	REGISTER Ry		

MULS

WORD															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	0	0	REGISTER			1	1	1	EFFECTIVE ADDRESS					
										MODE			REGISTER		

EXG

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	0	0	REGISTER Rx			1	OPMODE					REGISTER Ry		

AND

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	0	0	REGISTER			OPMODE			EFFECTIVE ADDRESS					
										MODE			REGISTER		

ADDX

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	0	1	REGISTER Rx			1	SIZE		0	0	R/M	REGISTER Ry		

ADDA

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	0	1	REGISTER			OPMODE			EFFECTIVE ADDRESS					
										MODE			REGISTER		



ADD

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	0	1	REGISTER			OPMODE			EFFECTIVE ADDRESS					
										MODE			REGISTER		

ASL, ASR

MEMORY SHIFT															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	0	0	0	0	dr	1	1	EFFECTIVE ADDRESS					
										MODE			REGISTER		

LSL, LSR

MEMORY SHIFT															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	0	0	0	1	dr	1	1	EFFECTIVE ADDRESS					
										MODE			REGISTER		

ROXL, ROXR

MEMORY ROTATE															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	0	0	1	0	dr	1	1	EFFECTIVE ADDRESS					
										MODE			REGISTER		

ROL, ROR

MEMORY ROTATE															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	0	0	1	1	dr	1	1	EFFECTIVE ADDRESS					
										MODE			REGISTER		

BFTST

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	0	1	0	0	0	1	1	EFFECTIVE ADDRESS					
										MODE			REGISTER		
0	0	0	0	Do	OFFSET					Dw	WIDTH				

BFEXTU

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	0	1	0	0	1	1	1	EFFECTIVE ADDRESS MODE REGISTER					
0	REGISTER			Do	OFFSET					Dw	WIDTH				

BFCHG

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	0	1	0	1	0	1	1	EFFECTIVE ADDRESS MODE REGISTER					
0	0	0	0	Do	OFFSET					Dw	WIDTH				

BFEXTS

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	0	1	0	1	1	1	1	EFFECTIVE ADDRESS MODE REGISTER					
0	REGISTER			Do	OFFSET					Dw	WIDTH				

BFCLR

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	0	1	1	0	0	1	1	EFFECTIVE ADDRESS MODE REGISTER					
0	0	0	0	Do	OFFSET					Dw	WIDTH				

BFFFO

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	0	1	1	0	1	1	1	EFFECTIVE ADDRESS MODE REGISTER					
0	REGISTER			Do	OFFSET					Dw	WIDTH				

BFSET

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	0	1	1	1	0	1	1	EFFECTIVE ADDRESS MODE REGISTER					
0	0	0	0	Do	OFFSET					Dw	WIDTH				



BFINS

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	0	1	1	1	1	1	1	EFFECTIVE ADDRESS					
										MODE		REGISTER			
0	REGISTER			Do	OFFSET					Dw	WIDTH				

ASL, ASR

REGISTER SHIFT															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	0	COUNT/ REGISTER		dr		SIZE		i/r	0	0	REGISTER		

LSL, LSR

REGISTER SHIFT															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	0	COUNT/ REGISTER		dr		SIZE		i/r	0	1	REGISTER		

ROXL, ROXR

REGISTER ROTATE															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	0	COUNT/ REGISTER		dr		SIZE		i/r	1	0	REGISTER		

ROL, ROR

REGISTER ROTATE															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	0	COUNT/ REGISTER		dr		SIZE		i/r	1	1	REGISTER		

PMOVE

MC68EC030, ACX REGISTERS															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	0	0	0	0	0	0	EFFECTIVE ADDRESS					
											MODE		REGISTER		
0	0	0	P REGISTER			R/W	0	0	0	0	0	0	0	0	0

PMOVE

MC68030 ONLY, TT REGISTERS

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	0	0	0	0	0	0	EFFECTIVE ADDRESS					
										MODE			REGISTER		
0	0	0	P REGISTER			R/W	FD	0	0	0	0	0	0	0	0

PLOAD

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	0	0	0	0	0	0	EFFECTIVE ADDRESS					
											MODE		REGISTER		
0	0	1	0	0	0	R/W	0	0	0	0	FC				

PVALID

VAL CONTAINS ACCESS LEVEL TO TEST AGAINST

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	0	0	0	0	0	0	EFFECTIVE ADDRESS					
										MODE			REGISTER		
0	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0

PVALID

MAIN PROCESSOR REGISTER CONTAINS ACCESS LEVEL TO TEST AGAINST

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	0	0	0	0	0	0	EFFECTIVE ADDRESS					
										MODE			REGISTER		
0	0	1	0	1	0	0	0	0	0	0	0	0	REGISTER		

PFLUSH

MC68030 ONLY

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	0	0	0	0	0	0	EFFECTIVE ADDRESS					
											MODE		REGISTER		
0	0	1	MODE			0	0	MASK			FC				

PFLUSH
PFLUSHA
PFLUSHS

MC68851															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	0	0	0	0	0	0	EFFECTIVE ADDRESS					
										MODE			REGISTER		
0	0	1	MODE			0	MASK				FC				

PMOVE

MC68851, TO/FROM TC, CRP, DRP, SRP, CAL, VAL, SCC, AND AC REGISTERS															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	0	0	0	0	0	0	EFFECTIVE ADDRESS					
										MODE			REGISTER		
0	1	0	P REGISTER			R/W	0	0	0	0	0	0	0	0	0

PMOVE

MC68030 ONLY, SRP, CRP, AND TC REGISTERS															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	0	0	0	0	0	0	EFFECTIVE ADDRESS					
										MODE			REGISTER		
0	1	0	P REGISTER			R/W	FD	0	0	0	0	0	0	0	0

PMOVE

MC68030 ONLY, MMUSR															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	0	0	0	0	0	0	EFFECTIVE ADDRESS					
										MODE			REGISTER		
0	1	1	0	0	0	R/W	0	0	0	0	0	0	0	0	0

PMOVE

MC68EC030, ACUSR															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	0	0	0	0	0	0	EFFECTIVE ADDRESS					
										MODE			REGISTER		
0	1	1	0	0	0	R/W	0	0	0	0	0	0	0	0	0

PMOVE

MC68851, TO/FROM PSR AND PCSR REGISTERS

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	0	0	0	0	0	0	EFFECTIVE ADDRESS					
										MODE			REGISTER		
0	1	1	P REGISTER			R/W	0	0	0	0	0	0	0	0	0

PMOVE

MC68851, TO/FROM BADX AND BACX REGISTERS

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	0	0	0	0	0	0	EFFECTIVE ADDRESS					
										MODE			REGISTER		
0	1	1	P REGISTER			R/W	0	0	0	0	NUM		0	0	0

PTEST

MC68EC030

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	0	0	0	0	0	0	EFFECTIVE ADDRESS					
										MODE			REGISTER		
1	0	0	0	0	0	R/W	0	REGISTER			FC				

PTEST

MC68030 ONLY

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	0	0	0	0	0	0	EFFECTIVE ADDRESS					
										MODE			REGISTER		
1	0	0	LEVEL			R/W	A	REGISTER		FC					

PTEST

MC68851

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	0	0	0	0	0	0	EFFECTIVE ADDRESS					
										MODE			REGISTER		
1	0	0	LEVEL			R/W	A REGISTER			FC					

PFLUSHR

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	0	0	0	0	0	0	EFFECTIVE ADDRESS					
										MODE		REGISTER			
1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0

PScc

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	0	0	0	0	0	1	EFFECTIVE ADDRESS					
										MODE		REGISTER			
0	0	0	0	0	0	0	0	0	0	MC68851 CONDITION					

PDBcc

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	0	0	0	0	0	1	0	0	1	COUNT REGISTER		
0	0	0	0	0	0	0	0	0	0	MC68851 CONDITION					
16-BIT DISPLACEMENT															

PTRAPcc

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	0	0	0	0	0	1	1	1	1	OPMODE		
0	0	0	0	0	0	0	0	0	0	MC68851 CONDITION					
16-BIT OPERAND OR MOST SIGNIFICANT WORD OF 32-BIT OPERAND (IF NEEDED)															
LEAST SIGNIFICANT WORD OF 32-BIT OPERAND (IF NEEDED)															

PBcc

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	0	0	0	0	1	SIZE	MC68851 CONDITION					
16-BIT DISPLACEMENT OR MOST SIGNIFICANT WORD OF 32-BIT DISPLACEMENT															
LEAST SIGNIFICANT WORD OF 32-BIT DISPLACEMENT (IF NEEDED)															

PSAVE

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	0	0	0	1	0	0	EFFECTIVE ADDRESS					
										MODE		REGISTER			

PRESTORE

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	0	0	0	1	0	1	EFFECTIVE ADDRESS MODE REGISTER					

PFLUSH

MC68EC040, POSTINCREMENT SOURCE AND DESTINATION

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	0	1	0	1	0	0	0	OPMODE		REGISTER		

PFLUSH

MC68040/MC68LC040

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	0	1	0	1	0	0	0	OPMODE		REGISTER		

PTEST

MC68040/MC68LC040

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	0	1	0	1	0	1	R/W	0	1	REGISTER		

PTEST

MC68EC040

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	0	1	0	1	0	1	R/W	0	1	REGISTER		

CINV

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	0	1	0	0	CACHE		0	SCOPE		REGISTER		

CPUSH

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	0	1	0	0	CACHE		1	SCOPE		REGISTER		

MOVE16

ABSOLUTE LONG ADDRESS SOURCE OR DESTINATION															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	0	1	1	0	0	0	0	OPMODE		REGISTER Ay		
HIGH-ORDER ADDRESS															
LOW-ORDER ADDRESS															

MOVE16

POSTINCREMENT SOURCE AND DESTINATION															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	0	1	1	0	0	0	1	0	0	REGISTER Ax		
1	REGISTER Ay			0	0	0	0	0	0	0	0	0	0	0	0

TBLU, TBLUN

TABLE LOOKUP AND INTERPOLATE															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	1	0	0	0	0	0	EFFECTIVE ADDRESS					
					MODE			REGISTER							
0	REGISTER Dx			0	R	0	1	SIZE	0	0	0	0	0	0	0

TBLS, TBLSN

TABLE LOOKUP AND INTERPOLATE															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	1	0	0	0	0	0	EFFECTIVE ADDRESS					
					MODE			REGISTER							
0	REGISTER Dx			1	R	0	1	SIZE	0	0	0	0	0	0	0

TBLU, TBLUN

DATA REGISTER INTERPOLATE															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	1	0	0	0	0	0	0	0	0	REGISTER Dym		
0	REGISTER Dx			0	R	0	0	SIZE		0	0	0	REGISTER Dyn		

TBLS, TBLSN

DATA REGISTER INTERPOLATE

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	1	0	0	0	0	0	0	0	0	REGISTER Dym		
0	REGISTER Dx			1	R	0	0	SIZE			0	0	0	REGISTER Dyn	

LPSTOP

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	1	1	1	0	0	0	0	0	0
IMMEDIATE DATA															

FMOVECR

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	0	0	0	0	0	0
0	1	0	1	1	1	DESTINATION REGISTER			ROM OFFSET						

FINT

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS					
										MODE	REGISTER				
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			0	0	0	0	0	0	1

FSINH

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS					
										MODE	REGISTER				
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			0	0	0	0	0	1	0

FINTRZ

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS					
										MODE	REGISTER				
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			0	0	0	0	0	1	1



FLOGNP1

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS					
										MODE	REGISTER				
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			0	0	0	0	1	1	0

FETOXM1

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS					
										MODE	REGISTER				
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			0	0	0	1	0	0	0

FTANH

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS					
										MODE	REGISTER				
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			0	0	0	1	0	0	1

FATAN

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS					
										MODE	REGISTER				
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			0	0	0	1	0	1	0

FASIN

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS					
										MODE	REGISTER				
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			0	0	0	1	1	0	0

FATANH

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS					
										MODE		REGISTER			
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			0	0	0	1	1	0	1

FSIN

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID		0	0	0	EFFECTIVE ADDRESS						
										MODE		REGISTER			
0	R/M	0	SOURCE SPECIFIER		DESTINATION REGISTER			0	0	0	1	1	1	0	

FTAN

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID		0	0	0	EFFECTIVE ADDRESS						
										MODE		REGISTER			
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			0	0	0	1	1	1	1

FETOX

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID		0	0	0	EFFECTIVE ADDRESS						
										MODE		REGISTER			
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			0	0	1	0	0	0	0

FTWOTOX

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID		0	0	0	EFFECTIVE ADDRESS						
										MODE		REGISTER			
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			0	0	1	0	0	0	1

FTENTOX

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID		0	0	0	EFFECTIVE ADDRESS						
										MODE		REGISTER			
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			0	0	1	0	0	1	0

FLOGN

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS					
										MODE	REGISTER				
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			0	0	1	0	1	0	0

FLOG10

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS					
										MODE	REGISTER				
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			0	0	1	0	1	0	1

FLOG2

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS					
										MODE	REGISTER				
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			0	0	1	0	1	1	0

FCOSH

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS					
										MODE	REGISTER				
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			0	0	1	1	0	0	1

FACOS

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS					
										MODE	REGISTER				
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			0	0	1	1	1	0	0

FCOS

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS					
										MODE	REGISTER				
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			0	0	1	1	1	0	1

FGETEXP

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID		0	0	0	EFFECTIVE ADDRESS						
											MODE	REGISTER			
0	R/M	0	SOURCE SPECIFIER		DESTINATION REGISTER			0	0	1	1	1	1	0	

FGETMAN

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID		0	0	0	EFFECTIVE ADDRESS						
											MODE	REGISTER			
0	R/M	0	SOURCE SPECIFIER		DESTINATION REGISTER			0	0	1	1	1	1	1	

FMOD

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID		0	0	0	EFFECTIVE ADDRESS						
											MODE	REGISTER			
0	R/M	0	SOURCE SPECIFIER		DESTINATION REGISTER			0	1	0	0	0	0	0	1

FSGLDIV

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID		0	0	0	EFFECTIVE ADDRESS						
											MODE	REGISTER			
0	R/M	0	SOURCE SPECIFIER		DESTINATION REGISTER			0	1	0	0	1	0	0	

FREM

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID		0	0	0	EFFECTIVE ADDRESS						
											MODE	REGISTER			
0	R/M	0	SOURCE SPECIFIER		DESTINATION REGISTER			0	1	0	0	1	0	1	

FSCALE

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID		0	0	0	EFFECTIVE ADDRESS						
											MODE	REGISTER			
0	R/M	0	SOURCE SPECIFIER		DESTINATION REGISTER			0	1	0	0	1	1	0	



FSGLMUL

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS					
										MODE	REGISTER				
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			0	1	0	0	1	1	1

FSINCOS

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS					
										MODE	REGISTER				
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			0	1	1	0	DESTINATION REGISTER, FPC		

FCMP

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS					
										MODE	REGISTER				
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			0	1	1	1	0	0	0

FTST

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS					
										MODE	REGISTER				
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			0	1	1	1	0	1	0

FABS

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS MODE REGISTER					
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			OPMODE						

FADD

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS					
										MODE	REGISTER				
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			OPMODE						

FDIV

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS MODEREGISTER					
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			OPMODE						

FMOVE

DATA REGISTER, EFFECTIVE ADDRESS TO REGISTER

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS MODEREGISTER					
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			OPMODE						

FMUL

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS MODE REGISTER					
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			OPMODE						

FNEG

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS MODE REGISTER					
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			OPMODE						

FSQRT

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS MODE REGISTER					
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			OPMODE						

FSUB

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS MODE REGISTER					
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			OPMODE						



FMOVE

DATA REGISTER, REGISTER TO MEMORY

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	0	0	1	0	0	0	EFFECTIVE ADDRESS					
										MODE				REGISTER	
0	1	1	SOURCE SPECIFIER			DESTINATION REGISTER			K-FACTOR (IF REQUIRED)						

FMOVE

SYSTEM CONTROL REGISTER

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS					
										MODE				REGISTER	
1	0	dr	REGISTER SELECT			0	0	0	0	0	0	0	0	0	0

FMOVEM

CONTROL REGISTERS

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS					
										MODE				REGISTER	
1	0	dr	REGISTER SELECT			0	0	0	0	0	0	0	0	0	0

FMOVEM

DATA REGISTERS

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS					
										MODE				REGISTER	
1	1	dr	MODE			0	0	0	REGISTER LIST						

cpGEN

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS					
										MODE				REGISTER	
COPROCESSOR ID-DEPENDENT COMMAND WORD															
OPTIONAL EFFECTIVE ADDRESS OR COPROCESSOR ID-DEFINED EXTENSION WORDS															

FScc

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	1	EFFECTIVE ADDRESS MODE REGISTER					
0	0	0	0	0	0	0	0	0	0	CONDITIONAL PREDICATE					

cpScc

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	1	EFFECTIVE ADDRESS					
										MODE	REGISTER				
0	0	0	0	0	0	0	0	0	0	COPROCESSOR ID CONDITION					
OPTIONAL EFFECTIVE ADDRESS OR COPROCESSOR ID-DEFINED EXTENSION WORDS															

FBcc

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	1	SIZE	CONDITIONAL PREDICATE					
16-BIT DISPLACEMENT OR MOST SIGNIFICANT WORD OF 32-BIT DISPLACEMENT															
LEAST SIGNIFICANT WORD OF 32-BIT DISPLACEMENT (IF NEEDED)															

cpBcc

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	1	SIZE	COPROCESSOR ID CONDITION					
OPTIONAL COPROCESSOR ID-DEFINED EXTENSION WORDS															
WORD OR															
LONG-WORD DISPLACEMENT															

cpSAVE

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			1	0	0	EFFECTIVE ADDRESS MODE REGISTER					

FSAVE

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			1	0	0	EFFECTIVE ADDRESS MODE REGISTER					

cpRESTORE

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			1	0	1	EFFECTIVE ADDRESS					
										MODE			REGISTER		

FRESTORE

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			1	0	1	EFFECTIVE ADDRESS					
										MODE			REGISTER		

FDBcc

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	1	0	0	1	COUNT REGISTER		
0	0	0	0	0	0	0	0	0	0	CONDITIONAL PREDICATE					
16-BIT DISPLACEMENT															

cpDBcc

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	1	0	0	1	REGISTER		
0	0	0	0	0	0	0	0	0	0	COPROCESSOR ID CONDITION					
OPTIONAL COPROCESSOR ID-DEFINED EXTENSION WORDS															
16-BIT DISPLACEMENT															

FTRAPcc

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	1	1	1	1	MODE		
0	0	0	0	0	0	0	0	0	0	CONDITIONAL PREDICATE					
16-BIT OPERAND OR MOST SIGNIFICANT WORD OF 32-BIT OPERAND (IF NEEDED)															
LEAST SIGNIFICANT WORD OR 32-BIT OPERAND (IF NEEDED)															

cpTRAPcc

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	1	1	1	1	OPMODE		
0	0	0	0	0	0	0	0	0	0	COPROCESSOR ID CONDITION					
OPTIONAL COPROCESSOR ID-DEFINED EXTENSION WORDS															
OPTIONAL WORD															
OR LONG-WORD OPERAND															

FNOP

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	1	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

